

Engagement Document Set

PREPARED BY Energy Hippo implementation team

How to turn the internal specs/YAMLs into the documents a **real implementation engagement** actually exchanges — so Northlake reads like a genuine customer, with the paperwork a corporate onboarding would have.

Packet 0.3 is available: the [customer packet](#) now includes the corrected 26-tab workbook, complete facilities/meter register, source measurement mapping and first AcquiSuite files. The matching [UI implementation checklist](#) and [gateway coverage](#) are our response. The document catalog below describes the larger engagement; it is not a list of new prerequisites before starting the first gateway.

The key idea: it's two "binders," not one document type

What you're describing ("integration docs? ADRs? system design docs?") isn't a single artifact — it's a **document set** split across two sides:

- 1. Customer Intake Package** — what Northlake sends us. Business facts in the customer's own language: facilities, utility accounts, tariffs, bills, org chart, GL chart, leases, sustainability goals, data feeds. The customer does **not** send EEM configuration — they send the **requirements and source data**.
- 2. Solution & Integration Documents** — what we produce in response. The design and configuration: solution design, integration specs, decision records, build workbook, test plan, project plan.

Our YAMLs are mostly **side 2** (our design/config). The "like the customer sent it" docs are **side 1** — the **business facts that justify** each YAML. Example: the rates YAML (determinants) is our response to the customer's **rate tariff sheets**; the security YAML is our response to their **org chart + access request**; the tenant YAML is our response to their **lease roster**.

This already matches the repo: documents/ (side 1, with explicit "customer language vs implementation language" rules) vs data/eem/ + implementation/ (side 2). We extend both.

What each artifact is actually called

You said...	Industry name	Side
"like the customer sent it to us"	Implementation Intake / Discovery Package (incl. a Data Collection Workbook)	Customer
"integration docs"	Integration Specification / Interface Control Document (ICD) — one per data feed	Vendor
"system design docs"	Solution Design Document (SDD) (a.k.a. System Design Doc / Solution Blueprint)	Vendor
"ADRs"	Architecture Decision Records — one per discrete decision	Vendor
"plans"	Statement of Work (SOW) + Project/Implementation Plan	Vendor

Side 1 — Customer Intake Package: how Northlake tells us each thing

What a real customer hands over, by subsystem. Author = the customer department that would own it. (Voice + allowed vocabulary per `generators/documents/README.md`.)

Subsystem	Customer artifact	Author (Northlake)	Format	Status
Org & scope	Discovery questionnaire (completed) + cover letter	Project sponsor	Word/PDF	new
Facilities/hierarchy	Facilities & Utility Overview	Facilities	PDF	✓ have
Everything structured	Data Collection Workbook (inventory, rates, measurement relationships, source mapping, events and later requirements)	Facilities/Finance	Excel	Corrected, packet 0.3
Utility accounts	Utility account inventory (tab) + sample bills	Finance	Excel + PDF	✓ have
Rates	Rate tariff sheets per provider	Finance / utility	PDF	new
AP/GL	Chart of Accounts + GL coding guide ; AP vendor setup	Finance/AP	PDF/Excel	new
Users/security	Org chart + user access request / role matrix	HR/IT	Excel/PDF	new
Tenant rebilling	Tenant & lease roster + allocation-method memo	Housing/Property Mgmt	Excel/PDF	new
Sustainability	Reporting requirements (ENERGY STAR, GHG scopes, targets)	Sustainability	PDF	partial (in overview)
Projects	Capital/energy project list	Facilities/Sustainability	Excel	new
Data feeds	Source-system inventory + delivery cadence + sample files	IT/Facilities	Excel/Markdown/files	Complete mapping; AcquiSuite samples supplied; other fixtures pending

The Data Collection Workbook is the structured handover. Its 26 tabs are 21 generated tabs (Instructions, Hierarchy, Contacts, Departments & Responsibilities, Known Events, Sites, Buildings, Service Profiles, Utility Services, Rate Schedules, Rate Components, Accounts And Agreements, Meters, Units and Submeters, Measured Points, Related Measurements, Rollup Members, Weather Assignments, Open Items, Data Sources, Source Measurements) plus 5 retained later-phase tabs (Chart of Accounts, Users & Access, Tenants & Leases, Sustainability, Projects). Contacts and Departments & Responsibilities provide the complete implementation directory and department ownership. Hierarchy lists the company, site and building nodes (every company uses the same four levels: Site, Building, Meter, Point), with full meter parent paths on Meters; Service Profiles and Units and Submeters are new in packet 0.3. Separate inventory, source-system and tenant workbooks are retired. After `python generators/northlake_packet.py`, `python generators/update_workbook.py` rebuilds the generated tabs and leaves the later-phase tabs as they are. The [generator ownership table](#) identifies the versioned source for each field so the workbooks and Markdown registers stay consistent.

Side 2 — Solution & Integration Documents: what we produce

Document	Purpose	Built from	Format	Status
Statement of Work (SOW)	Scope, deliverables, phases, timeline, assumptions	roadmap.pdf	Word/PDF	derivable
Solution Design Document (SDD)	The EEM configuration design across all subsystems	subsystem-gap-analysis.pdf + phase specs	Word/PDF	derivable
Integration Specs / ICDs (one per feed)	Each interface: gateways, bill import, AP export, ENERGY STAR, EnergyAI	data/providers/ + gateway bindings + phase3/5/6 specs	Word/PDF	new
Data Mapping Specification	Customer workbook fields → EEM config fields	workbook + data/eem/ entry plans	Excel/Word	new
Configuration / Build Workbook	Step-by-step build (the "build book")	data/eem/*.yaml entry plans	Word/Excel	✓ have (as YAML)
Architecture Decision Records	One per design decision	decisions made across this project	Markdown/PDF	new (list below)
Test / UAT Plan	Acceptance criteria + expected results	phase-0-1-assertions.pdf + each plan's expected Assertions	Word/Excel	derivable
Gap & Risk Register (RAID)	Open product gaps + risks	product-gap-backlog.pdf	Excel/Word	✓ have
Project / Implementation Plan	Schedule, milestones, RACI	roadmap.pdf	Word/MS-Project-ish	derivable
Cutover / Go-Live Runbook	Load order + go-live steps	roadmap dependency order	Word	new

Candidate ADRs (decisions already made — ready to write up)

1. Model the central plant as **internal cost allocation**, not a utility account.
2. **Cedar Row** as a separate EEM company (security boundary).
3. Use the built-in **System company (CompanyID -1)** for implementation/support.
4. **Enter all data through the EEM UI / legitimate data paths — no direct DB seed.**
5. **Close UI gaps by building product features** (vs deferring).
6. Model rates at the **determinant level** to exercise all 9 determinant kinds.
7. **Connect a sandbox ENERGY STAR PM** account for scores (no hand-entry).
8. **Both Town Center + Cedar Row sub-metering** for tenant rebilling.
9. **Source-model-first**; numeric tariffs single-sourced in data/providers/.
10. **Deterministic synthetic identities**; public context only.

Repository structure

documents/	# Side 1: Northlake's own files (PDF)
intake-package/	# transmittal, questionnaire, binder, workbook (26 tabs),
	# facilities and meter register, source-system inventory,
	# user access request, AcquiSuite sample delivery
guides/	# facilities & utility overview, GL coding guide,
	# sustainability requirements, tenant & lease roster
rate-tariffs/<provider>.pdf	
utility-bills/<provider>/	# bills and allocation statements
brand/	# EEM portal brand kit
implementation/	# Side 2: what we produce (PDF)
dream-customer/	# coverage plan, roadmap, phase specs
briefs/	# design briefs
(planned) sow, solution-design-document, integration-specs/<feed>,	
data-mapping-specification, adr/NNNN-<decision>, test-uat-plan,	
gap-risk-register (from product-gap-backlog), implementation-plan	
(from roadmap), cutover-runbook	
data/	# the versioned YAML facts both sides are built from
generators/documents/sources/	# the Markdown behind every PDF above

Markdown stays the versionable source in `generators/documents/sources/`, and `python generators/documents/build_documents.py` renders each file as a Northlake PDF under `documents/` or `implementation/` for the "this is what they sent / what we delivered" feel. `python generators/update_workbook.py` maintains the one Excel workbook in `documents/intake-package/`. Skills available: docx, pdf, xlsx, pptx, plus engineering:architecture for ADRs.

Recommended sequencing

- 1. Customer Intake Package** first — it's the most evocative "real customer" artifact and becomes the **input** the Side-2 docs respond to. Start with the workbook expansion + the new intake docs (rate tariff sheets, GL guide, access request, lease roster, sustainability requirements, source-system inventory), plus a cover letter + discovery questionnaire.
- 2. Solution Design Document + ADRs** — the headline Side-2 docs that tie the intake to our config.
- 3. Integration Specs / ICDs, Data Mapping, Test/UAT Plan, SOW / Plan, Runbook** — fill out the binder.

Everything is a reframing of work already done — no new design, just the corporate packaging.